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Political connections, financing and firm performance: Evidence from Chinese private firms ☆

Hongbin Li [a,b,*], Lingsheng Meng [c], Qian Wang [d], Li-An Zhou [e]

[a] *Department of Economics, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong, PR China*
[b] *School of Economics and Management, Tsinghua University, Beijing 100084, PR China*
[c] *Department of Economics, University of Maryland, College Park, MD 20742, USA*
[d] *School of Accountancy, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong*
[e] *Guanghua School of Management, Peking University, Center for Research in Private Enterprise at Zhejiang University, PR China*

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Abstract

This paper examines the role of affiliation with the ruling Communist Party in the operation of private enterprises in China. Using a nationwide survey of private firms, we find that the Party membership of private entrepreneurs has a positive effect on the performance of their firms when human capital and other relevant variables are controlled. We further find that Party membership helps private entrepreneurs to obtain loans from banks or other state institutions, and affords them more confidence in the legal system. Finally, we find Party membership to be more important to firm performance in regions with weaker market institutions and weaker legal protection.
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1. Introduction

There is a growing body of economic literature on the implications of political connections – and in particular the measurement of the value of political connections – in the business world. It has been found that political connections help firms to secure favorable regulatory conditions (Agrawal and Knoeber, 2001) and access to resources such as bank loans (Khwaja and Mian, 2005; Faccio, 2006), which ultimately increases the value of firms (Roberts, 1990; Fisman, 2001; Ramalho, 2007) or improves their performance (Johnson and Mitton, 2003).

The cultivation of political connections is also important for private entrepreneurs in transition economies such as that of China. The incentive for businessmen to establish political connections in transition countries ultimately arises from the state control of key resources. Due to the lingering legacy of the command economy and the slow development of market-supporting institutions, private entrepreneurs in transition economies face many

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* Corresponding author. Tel.: +852 2609 8185; fax: +852 2603 5805.
E-mail address: lhongbin@cuhk.edu.hk (H. Li).

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obstacles in running their businesses. They are often denied access to bank loans, which are largely reserved for state-owned enterprises, or are subject to heavy government regulations (red tape) or "extralegal" fees (Johnson et al., 2000; McMillan and Woodruff, 2002; Guriev, 2004). In addition to the problems of a weak state and ill-functioning markets, the legal system in such economies is often too weak to secure property rights and enforce contracts (Hay and Shleifer, 1998; McMillan and Woodruff, 1999; Frye and Zhuravskaia, 2000). In such an environment, close ties to the governments help businesses to overcome these market and state failures and avoid ideological discrimination.

In this paper, we examine the value of political connections for private entrepreneurs in China. In particular, we analyze the role of affiliation with the Chinese Communist Party (the Party) in helping to improve the performance of private firms, and investigate the specific mechanisms through which this political affiliation operates. A distinct feature of China that sets it apart from other transition countries is the continuing rule of the Communist Party and the continuing ideological discrimination against private ownership, despite the dramatic reform of the economy. Party membership in China implies a measure of political status, and, more importantly, represents an important affiliation with the only ruling party in China. In an environment in which skepticism, mistrust, and discrimination toward private enterprises abound, affiliation with the ruling party confers congruence and trustworthiness. As Party members interact with government officials, bank managers, and managers of state-owned enterprises, they build up connections with key political and economic figures. Furthermore, being a Party member and becoming acquainted with other Party members and government officials create an avenue for the achievement of a higher political status, such as becoming a member of the People's Congress (Li et al., 2006).

Our first empirical task is to examine whether Communist Party membership has a positive effect on the profitability of private firms. We do so by using a unique nationwide survey of more than 2000 private firms in China, and find that the Party member status of the owner contributes significantly to a firm's profitability. This finding is robust to a series of alternative specifications that control for the characteristics of entrepreneurs and enterprises. The residual power of Party membership to explain firm performance after controlling for the characteristics of entrepreneurs suggests that it is indeed a driver of firm profitability.

We then go a step further and investigate how and where Party membership is important to firm performance. We argue that in a transition economy such as that of China, the role of political connections may be more important than it would be in a mature economy, because of certain institutional features. In the case of China, close ties with the government or ruling party help private enterprises to overcome legal and institutional failure and ideological discrimination against private ownership. Our empirical results show that political connections affect firm performance through a number of mechanisms that are related to the weak institutional environment in China. In particular, we find that Party membership status helps entrepreneurs to obtain loans from banks or other state institutions, and that politically connected firms are more likely to resort to the courts in business disputes, which suggests that they have more confidence in the legal system. Finally, we find that the effect of Party membership on firm performance is more important in regions with less developed markets and legal systems. These findings support the view that political connections can significantly help firms to overcome the market failures that are prevalent in China.

Although our focus is to test whether Party membership is an important political connection that affects firm performance, our results are also consistent with the hypothesis that Party membership is an indicator of the human capital or ability of an entrepreneur, because the strict and merit-based selection process of the Party means that its members comprise China's elite. Although we are not able to discount the human capital hypothesis completely, we do find some evidence to differentiate it from our political connections hypothesis. If Party membership represents human capital, then one would expect it to be more valuable in more developed market environments. However, we find on the contrary that Party membership is more important when the market is less developed, which seems to suggest that Party membership probably leads to the establishment of political connections that are more significant in a weak market environment.

Our paper adds further evidence to the growing literature on the implications of political connections, and in particular on the measurement of the value of political connections in the business world. It is also among the first to examine the role of political connections in a transition economy.[1] Morduch and Sicular (2000) find that Party

[1] Our paper is also related to Allen et al. (2005) who argue that the rise of China's private sector is a counterexample to prevailing thought in the literature on law, institutions, finance, and growth that stresses the role of legal origin and institutions in finance and growth (La Porta et al., 1998, 1999, 2000; Demirguc-Kunt and Maksimovic, 1998; Levine, 1999; Beck and Levine, 2002; Djankov et al., 2003). According to Allen et al., the key to the development of the private sector in China is personal relationships and the active role of local government.

members or local cadres enjoyed a considerable amount of political rent in rural China in the 1990s, and Liu (2003) provides evidence that Party membership had a substantial effect on pecuniary and non-pecuniary income in China in the 1980s. Li et al. (2006) find that private entrepreneurs in China enter politics to overcome certain institutional difficulties, and Bai et al. (2005) find that political connections help private firms to obtain bank loans. Hellman et al. (2003) find that in high-capture economies, captor firms bribe public cadres in return for property rights and to secure superior firm performance. However, whereas Hellman et al. emphasize the role of corruption, we stress political connections, rather than corruption, as a source of state capture and influence. In a more recent work, Fan et al. (in press) find that political connections may lead to the poor corporate governance of listed state-owned enterprises in China, a somewhat unsurprising finding given that the managers of listed state-owned enterprises – and especially those with good government connections – have incentives other than profit maximization (e.g., political and social objectives). We find that political connections help private firms because the most important goal of a private entrepreneur is to maximize profit, and private firms in China have long been economically and ideologically discriminated against by the government.

The structure of the remainder of this paper is as follows. Section 2 introduces the institutional background to private enterprises in China and the role of political connections. Section 3 proposes hypotheses on the way in which political connections and human capital relate to the Party membership of enterprise owners, and outlines the econometric specification. Section 4 describes the data and variables. Sections 5 and 6 empirically test the hypotheses and report the results, and Section 7 provides a conclusion.

2. Institutional background

2.1. The evolving status of private business in China

In this subsection, we briefly describe the development of private enterprises in the history of the People's Republic of China (PRC), with particular focus on the institutional environment in which private firms have grown up in the past two and a half decades. These institutional details should aid our understanding of the value of political connections to private enterprises.

When the Communist Party (henceforth the Party) won the civil war and founded the PRC in 1949, it began the socialist transformation of private enterprises. Between 1952 and 1977 private enterprises were completely banned in China, and although they were accorded a measure of political tolerance when they resurfaced in the early 1980s,[2] they were not allowed to hire more than eight employees. This rule remained in place for ten years, after which time the National People's Congress authorized the establishment of private enterprises with more than eight employees.[3]

Since Deng Xiaoping's Southern Tour of 1992, private business has advanced in leaps and bounds. China's private sector grew from nothing in the late 1970s to providing nearly 50% of the total employment and 60% of the industrial output by 2004. Since its revival, the growth rate of the private sector has far outpaced that of the public sector. One of the most significant changes in the development of the private sector since 1978 occurred in March 2004, when the National People's Congress approved a constitutional amendment to protect private property rights, marking the first time in PRC history that the legal status of private property was officially endorsed by the Party.

Despite the speed with which the private sector developed after 1978, private firms have suffered both political and social discrimination.[4] Even in the late 1990s, private firms were still considered to be an inferior form of ownership for ideological reasons, and despite the existence of formal legislation that permitted private enterprises, the overall political environment was antagonistic toward the private sector. Private entrepreneurs had to deal with hostility and social prejudice on the part of cadres and the public in general, who regarded them to be dubious, ignoble, and even despicable. Challenges to the legitimacy of private enterprises also came from various political movements, such as the periodic campaigns against spiritual pollution in 1983 and 1984 and against bourgeois liberalization in 1987, and other movements that demanded a crackdown on private enterprises on the basis of "rectifying the market" or "attacking speculation."

Private firms in China not only experience political and social discrimination, but must also deal with an unfavorable economic environment. The government still controls most of the resources, and state-owned enterprises still enjoy preferential status in obtaining bank loans and other key inputs (Che, 2002; Brandt and

[2] They were presented as peripheral in the Constitution of 1978, and were considered to be "a marginal sector to be tolerated temporarily and tightly controlled" (Young, 1995, p. 14).

[3] An edict entitled "Provisional Regulations of the People's Republic of China on Private Enterprises" was issued in June 1988. See the People's Daily (Renmin Ribao), June 29, 1988.

[4] See Young (1995) for a comprehensive description of the development of the private sector in China since 1978.

Li, 2003). Private enterprises are subject to arbitrary harassment by government cadres (Pearson, 1997), and commercial and property laws are either non-existent or unenforceable (McMillan, 1995). Even today, despite the constitutional amendment in 2004 protecting private property rights, the Party is struggling to create fair market conditions such that private firms can compete with firms with different forms of ownership. There is reason to believe that it will be a long time before private firms acquire equal status with other types of firms, such as state-owned enterprises and foreign-funded firms.

2.2. The attainment of Party membership and "red capitalists"

The attainment of Party membership follows a distinct selection process that generally involves five stages: self-selection, political participation, daily monitoring, closed-door evaluation, and probationary examination (Bian et al., 2001). The process begins with an adult individual filing a formal application to a Party branch in their work unit to express their desire to become a member. The applicant is then monitored daily for at least three years, during which time they must make a continuous effort to meet all of the Party standards. Each applicant is assigned two Party members as liaisons who regularly monitor and assess the applicant's political loyalty, work performance, and collaboration with co-workers. When the Party branch believes that it is time to make a more thorough evaluation, usually about two years after the initial application, it seeks opinions about the applicant from non-Party co-workers and then has a closed-door evaluation that involves all of the Party members in the branch. Any serious doubt on the part of either a non-Party co-worker or a Party member can result in failure, and the applicant will be given time to improve before being considered for another closed-door evaluation. If the potential candidate passes the closed-door evaluation, then they become a probationary Party member, and will then be closely monitored by the Party branch for another year before being afforded full membership.

This lengthy and extended selection process ensures not only the political loyalty of applicants, but also the superior quality of Party members. To become a Party member, a person needs to show great ability by outperforming co-workers, good interpersonal skills by maintaining good relationships with co-workers, great persistence by performing well during the long selection process, and a positive attitude toward society, work, and the communist ideology. As work ability, interpersonal skills, persistence, and a positive attitude are also qualities that are important for the success of an individual generally in their social life and career, some sociologists (Szelenyi, 1987; Lin and Bian, 1991; Walder, 1995) view Party membership as a credential much like educational attainment. An important change in the selection criteria of Party members since economic reform in the late 1970s is the declining role of family class origin and the increasing emphasis on the educational credentials and expertise of applicants (Bian et al., 2001). As a result of this adjustment, Party members are now younger, better educated, and more likely to be engaged in knowledge-intensive occupations (such as college teachers, researchers, and engineers) than in the past.

It took a long time for the Party to encourage and accept applications for membership from the pool of private business owners. Originally, the Party claimed to represent the working class of poor peasants and workers, and thus did not extend its membership to private entrepreneurs, who had long been regarded as the economic class of capitalists. However, after the initiation of economic reform, the Party gradually liberalized its tight control over society and downplayed its original ideology. However, the criteria that private entrepreneurs had to fulfill to join the Party were still strict, and successful cases were rare. A major breakthrough occurred at the Party's Sixteenth Congress in 2002, when the membership charter was revised and private entrepreneurs were formally allowed to apply for Party membership.

The change in political sentiment toward private firms also encouraged many Party members to start their own businesses. As the private sector expended dramatically following Deng Xiaoping's Southern Tour of 1992, increasing numbers of Party members and government employees quit their Party or government posts to enter the private sector. This exodus from the government and government-controlled institutions to private firms has been vividly dubbed as "plunge into the sea" (*xiahai*). These new entrants to the thriving private sector have become an important contingent of what are popularly known as the "red capitalists" in China, that is, entrepreneurs with close personal and political ties to the Party.[5]

A number of factors contributed to the defection of government staff and managers of state-owned enterprises to the private sector after 1992. Deng Xiaoping's Southern Tour of 1992 represented a critical turnaround in the Party's policy toward the development of private

[5] See Dickson (2003) for a detailed analysis of the implications of "xiahai " and "red capitalists" in China.

enterprises, and since then restrictions on private ownership have been gradually relaxed and the political and economic environment has become friendlier to private enterprises. The emergence and promising prospect of private business has thus become increasingly attractive to government-sector workers. A more important reason for the emergence of "*xiahai*" entrepreneurs is that their previous work experience in the government or state-owned enterprises allows them to establish important connections with key Party and government officials before starting up their own firms. Given the adverse political and economic environment, Party membership has become a valuable political tool for private entrepreneurs, and indeed, to strengthen links with the Party an increasing number of private firms have established Party branches and regularly undertake Party activities.

Connections with the Party and the government afford the "*xiahai*" entrepreneurs certain advantages over other private entrepreneurs without such connections. These advantages include better access to key resources that are controlled by the Party and the government, such as business operation licenses, bank loans, land, and eligibility for favorable but discretionary government policies such as tax benefits and the waiver of "extralegal" fees. Being a Party member and getting to know other Party members and government officials also create an avenue for the achievement of higher political status, such as becoming a member of the People's Congress (PC), through which connections with the Party and the government can be further strengthened. Li et al. (2006) present evidence that private entrepreneurs with Party membership are more likely to become members of the PC.

3. Hypotheses and empirical strategy

In this section, we describe the main hypothesis on the role of Party membership for private firms, which we term the political connections hypothesis. We also briefly discuss a major competing hypothesis, the human capital hypothesis, and design tests to differentiate the two.

3.1. The political connections hypothesis

As has been discussed, transition countries are characterized by the existence of several institutional difficulties for private enterprises. These difficulties impose a huge cost on private firms, and ultimately damage their performance. Political connections or status help to reduce this cost, and thus improve firm performance. This leads to our first hypothesis: in China, Party membership is an important political connection that enhances the performance of private firms.

We test the first hypothesis by estimating the profitability equations using return on assets (ROA) and return on equity (ROE) as the dependent variables. More specifically, we estimate the following performance equation.

$$\text{Profitability} = \beta_0 + \beta_1 P + X\delta + \varepsilon, \quad (1)$$

where profitability is measured by ROA and ROE, and P is the Party membership dummy. The vector X represents a complete set of control variables that include the human capital of entrepreneurs and other political connection measures, and industry and provincial dummies.

We hypothesize that firm profitability is larger when the firm owner is a Party member, that is, $\beta_1 > 0$.

Estimating Eq. (1) may not be sufficient to test the validity of the first hypothesis, because $\beta_1 > 0$ could also mean that Party members have greater ability due to the selection process of the Party, as explained in Section 2. Thus, in our empirical test of the political connections hypothesis, we must be aware of this competing human capital hypothesis and try to differentiate it from our political connections hypothesis.

3.2. Further hypotheses and tests

Because there are two competing hypotheses, the political connections hypothesis and the human capital hypothesis, we need to develop further hypotheses and tests to establish evidence of the role of political connections. We do so by examining the concrete mechanisms through which political connections exert a positive effect.

We first develop hypotheses on the importance of Party membership. Private entrepreneurs in transition economies face many obstacles as a result of the lingering legacy of the command economy and the slow development of market-supporting institutions, obstacles that have their roots in the government control of the economy.

Private firms are often denied access to bank loans, which are largely reserved for state-owned enterprises (Nee, 1992; McMillan, 1997; Brandt and Li, 2003), and indeed, access to bank loans is one of the key differences between the state-owned enterprises and privately owned enterprises. State banks were not allowed to make loans to private firms until 1997, and a credit market in which state banks are almost the only source of

bank credit in effect has meant virtual exclusion for private firms. Although the situation has improved since 1997, private firms are still treated unfavorably in state-dominated financial markets. As a result, most private entrepreneurs rely on self-financing to start and expand their firms.

Governments may also impose heavy regulations (red tape) or "extralegal" fees on private enterprises (De Soto, 1989; Brunetti et al., 1997; Johnson et al., 2000, 2002; Hellman et al., 2003; Guriev, 2004), and legal systems may be too weak to secure property rights and enforce contracts (Hay and Shleifer, 1998; McMillan and Woodruff, 1999; Frye and Zhuravskaia, 2000). These barriers are greater in China because of the hostility of the political environment to private enterprises. Close ties with the government or ruling Party help enterprises to overcome these market and state failures and avoid ideological discrimination. Chinese private entrepreneurs who are well connected with the Party and the government are more likely than those without such ties to be able to obtain favorable treatment, such as the securing of bank loans, the circumvention of government regulations, and the obtaining of tax deductions and legal or non-legal protection for their enterprises.[6] In accordance with this argument, we test whether the positive effect of political connections occurs through mechanisms such as gaining access to the credit market, the reduction of red tape or taxes and fees, and better access to the legal system.

We test this hypothesis by using Eq. (1) with different dependent variables. We first test whether political connections help a firm to gain access to the credit market by using the value of loans obtained from state banks and other state institutions as the dependent variable. Second, we test whether political connections help a firm to avoid paying high taxes and fees by using two dependent variables: the tax rate, which is defined as the total tax payment divided by sales, and the fee rate, which is defined as the total fee payment divided by sales.[7] Third, we test the red-tape hypothesis by examining whether politically connected entrepreneurs spend less time dealing with bureaucracy. Finally, we test whether Party membership gives private entrepreneurs more confidence in the legal system by testing whether Party members are more likely to resort to legal channels in business disputes.

We also examine the situations in which Party membership matters most. In particular, we hypothesize that political connections are more important in weaker institutional environments, because being a Party member can help entrepreneurs in the weak institutional environment of China to increase the profitability of their firms. However, the positive role of Party membership in this respect may depend on the level of market development and the legal effectiveness of a region. If affiliation with the Party helps private entrepreneurs by sheltering them from ideological discrimination and increasing their confidence in the state-controlled courts, then we would expect this positive role to diminish in regions with relatively developed markets and legal systems. Empirical evidence shows that institutions differ dramatically across China (Brandt and Li, 2003; Lu and Yao, 2004), and this variation in institutions provides us with an opportunity to test the institutional factors behind our first hypothesis. Simply put, our last hypothesis states that if Party membership does indeed make firms more profitable in regions with weak institutions, then this difference should decrease as the institutions improve.

To test the relationship between Party membership and institutional environment, we construct a number of institutional indices to measure institutional heterogeneity across regions. These indices can be divided into two categories. The first category includes two indices that measure the development of the market in a region in terms of the proportion of employment in a province that is in the private sector and in terms of the proportion of total fixed investment that comes from private enterprises. We argue that when the private sector is small or the state sector large, private firms are obliged to have business dealings with state-owned firms, and thus connections with the government become very valuable. Moreover, a smaller private sector in a province implies greater regulation of the provincial economy by the state, which also makes political connections more important in doing business.

The second category comprises a single index that measures the effectiveness of legal protection in a region. In a recent study, Lu and Yao (2004) propose measuring the rule of law in China's provinces by taking the number of cases received by the court weighted by the provincial population. As they argue, this index measures the trust that people have in the legal system. We follow their lead and use the weighted number of cases received by the court in a province in

[6] Likewise, in the United States, individuals and interest groups make political campaign contributions in anticipation of receiving "services" in return, which include tax exemptions and the alleviation of regulatory burdens. See Snyder (1990) for a review of the relevant literature.

[7] The largest tax for firms in China is the value-added tax, which is based on sales revenue (less materials cost). There is also a profit tax, but profit cannot be used as a denominator because it can be zero or negative.

2001 as a measure of the legal effectiveness in that province.[8]

Again, our test is executed by estimating Eq. (1) with the introduction of two sets of institutional indices and interaction terms between the institutional indices and Party membership. Our hypothesis is that the estimated coefficients on the interaction terms should be negative, or, in other words, that the positive role of Party membership becomes less pronounced with a more developed private sector or legal system.

4. Data

The firm data that we use in this study come from a nationwide survey of privately owned enterprises that was jointly conducted in 2002 by the All China Industry and Commerce Federation, the China Society of Private Economy at the Chinese Academy of Social Sciences, and the United Front Work Department of the Central Committee of the Communist Party of China. The sample comprises mainly large firms and a small portion of individual household enterprises that were drawn from 31 provinces that cover all of the political subdivisions at this level in mainland China.

The dataset is by far the best for studying the effect of political connections on the performance of private enterprises in China, and comprises 3258 privately owned enterprises, which represents 0.16% of the total number of privately owned enterprises nationwide. The survey on which the dataset is based involved intensive interviews with firm owners, who we define as entrepreneurs, with questions about the size, history, and basic financial background of their firms and their family background, human capital, and occupational history.[9] Most importantly, the survey collected information on the political connections of the entrepreneurs, such as Party membership, previous work experience as a manager of a public firm or as a government cadre, and membership of the People's Congress (PC) or the Chinese People's Political Consultative Conference (CPPCC). There are 2324 observations for which there is relatively complete information on the variables of interest.

A primary analysis of the data shows quite a number of the private entrepreneurs to be Party members, a full 26% in fact. Unsurprisingly, most of these entrepreneurs (95%) joined the Party before they started their own business, which makes us less worried about the problem of reverse causality that would arise if the entrepreneurs had joined the Party because of their success in businesses. The percentage of entrepreneurs with experience of managing a public firm or service as a government cadre is also high, at 25% and 39%, respectively. The data also show 19% of the entrepreneurs to hold membership of the PC at various levels, and 34% to be members of the CPPCC.

The work experience of the entrepreneurs is reported in great detail in the dataset. On average, they have more than 11 years of management experience (Table 1). Their firms fall into diverse industrial sectors, ranging from farming to technical services. The firms have an average of 166 employees, which is much larger than the national average of 11.4 (YICAC, 2003), although firm size varies greatly across the sample, ranging from individual household enterprises to large-scale enterprises with more than 3000 employees.

In our sample, Party member entrepreneurs are different from non-Party members, and the firms that are owned by Party members differ from those that are owned by non-Party members. Table 2 reports the characteristics of the entrepreneurs and firms for Party members and non-Party members, respectively, and the t-tests of the differences between the two groups. As shown in the last column of Table 2, Party members are older, more educated, and have less management experience, and are more likely to have served as managers of public firms or government cadres in the past. Moreover, they are more likely to be PC members and less likely to be CPPCC members. All of these differences are significant at the one-percent level. The firms that are owned by Party members are generally larger – whether measured by sales volume, employment, or total assets – older, and more leveraged (initially and at present). They also appear to have better performance as measured by ROA and ROE, but these differences are not significant. In general, this simple tabulation shows that the firm owners and their firms do indeed differ with Party membership status.

The data for constructing the three institutional indices, as introduced in the last section, were obtained from the Statistical Yearbook of China for 2002. Note that provincial market development varies greatly in China. The mean proportion of employment in the private sector is 0.16, with a standard deviation of 0.08. The smallest proportion is only 4% of the provincial employment, whereas the largest proportion is 32%.

[8] Lu and Yao (2004) also use the ratio of closed cases to all commercial court cases in a year as a measure for the effectiveness of the legal system. However, this measure shows little variation, because the Supreme Court has pushed very hard to urge regional courts to increase the ratio of closed cases since the mid-1990s. Thus, we do not use this measure in our study.

[9] See Table A1 for a detailed description of the variables.

Table 1
Summary statistics

Variables	Observations	Mean	Standard Deviation	Min	Max
Owner's attributes					
Party membership	2324	0.26	0.44	0	1
Education	2324	12.85	3.02	0	19
Age	2324	44.10	8.05	22	75
Management experience (years)	2324	11.37	5.99	0	36
Former public firm manager	2324	0.25	0.43	0	1
Former cadre	2324	0.39	0.49	0	1
PC membership	2324	0.19	0.39	0	1
CPPCC membership	2324	0.34	0.47	0	1
Firm's attributes					
Sales (million yuan)	2307	21.16	53.04	0	701.25
Employment (persons)	2324	165.90	293.71	0	3250
Total assets (million yuan)	2324	14.18	51.34	0.03	1282.37
Total equity (million yuan)	2324	11.07	47.82	0.01	1250
ROA	2324	0.17	0.75	−0.47	28.08
ROE	2324	0.24	0.83	−1.45	28.08
Firm age (years)	2324	7.04	4.32	0	27
Leverage	2324	0.22	0.24	0	0.996
Initial loan from banks (million yuan)	2324	0.58	3.40	0	80
Initial loan from government institutions (million yuan)	2324	0.05	0.51	0	10
Total initial loan (million yuan)	2324	0.63	3.49	0	80
Total current loan (million yuan)	2278	2.38	8.49	0	200
Value added (million yuan)	2270	3.46	7.14	−0.92	105.25
Tax/sales	2274	0.06	0.16	0	7.25
Fees/sales	2303	0.02	0.06	0	0.89
Time dealing with bureaucracies	2161	1.86	1.11	0	12
Trust in the legal system	2324	0.09	0.29	0	1
Institutional indices					
Market 1	2324	0.16	0.08	0.04	0.32
Market 2	2324	0.54	0.09	0.05	0.65
Legal effectiveness	2265	121.49	61.09	26.14	224.31

Similarly, there is also a large cross-province variation in the effectiveness of the legal system. Such variation is important if we are to link these factors to the value of Party membership.

5. Party membership and profitability

In this section, we examine whether Party membership affects firm profitability. We use two measures of firm profitability, namely, return on assets (ROA) and return on equity (ROE). We employ ordinary least-squares regressions for all of the equations, and report the standard errors in parenthesis. All of the regressions control for a complete set of provincial and industry dummies.

The key to the test is to differentiate the political connections hypothesis from the human capital hypothesis. Party membership may entail political connections that help entrepreneurs to relax the constraints of the production process and thus increase profitability, but entrepreneurs who are Party members may also possess better human capital, as people with good human capital are more likely to be selected by the Party to be members. Although it is generally very difficult to completely isolate the political connections element from the human capital element, the availability of several other human capital and political connection variables may help us to partially differentiate the two. To undertake this differentiation, we include in our regressions a number of human capital measures of private entrepreneurs, including their education, age, and management experience.

The regression results show Party membership to have a positive effect on firm profitability. We begin by estimating our benchmark performance equation with the Party membership dummy, total assets in

log,[10] leverage ratio, firm age, the manager's human capital attributes, and the provincial and industry dummies as independent variables. The estimation results are presented in the first column of Table 3. The Party membership dummy, which is the primary variable of interest, has a positive coefficient that is significant at the five-percent level, and the owner being a Party member increases a firm's ROA by 8 percentage points. The control variables for size and leverage are significant in the regression, with both a larger firm size (as measured by total assets) and higher leverage being associated with lower returns. Firm age is not an important predictor of profitability. Of all of the human capital variables, only the age of the owner is significant, and has a negative effect on ROA. As we control for the human capital variables of the owner, the finding that Party membership has a significant effect on firm performance is not likely to be caused by a correlation between Party membership and these human capital variables.

In our next test, we examine whether Party membership takes effect through other channels. We conduct this test by including a number of other political connection variables, including a dummy for being a former public firm manager, a dummy for being a former cadre, a dummy for being a current member of the PC, and a dummy for being a current member of the CPPCC. The regressions that are shown in columns 2 and 3 demonstrate that the positive effect of Party membership remains large even after the addition of these new variables. The coefficient of the Party membership dummy is positive and significant at the five-percent level in both cases, and the magnitude of the effect remains almost constant after we control for the other political connection variables. Surprisingly, these newly added political connection variables do not have a significant effect on firm performance.

We urge caution in the interpretation of the results for the PC and CPPCC membership dummies, as they differ somewhat from the other political connection variables in our analysis. All of the other political connection variables are determined before the entrepreneurs started up their enterprises, which helps us to avoid the problem of reverse causality. However, the fact that most of the entrepreneurs obtained their membership of the PC or CPPCC after they started their enterprises could cause

[10] We also experiment on using different measures and forms of firm size such as log employment, both log employment and log total assets, and employment and employment squared, with little effect on the results. See Table A3 in Appendix C for more details.

Table 2
Summary statistics: Party members versus non-members

Variables	Party members	Non-members	Difference
	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)
	(1)	(2)	(1) − (2)
Observations	593	1731	
Owner's attributes			
Education	13.18	12.73	0.45
	(0.13)	(0.07)	(0.14)***
Age	46.94	43.13	3.80
	(0.31)	(0.19)	(0.37)***
Management experience (years)	10.69	11.60	−0.91
	(0.28)	(0.14)	(0.29)***
Former public firm manager	0.44	0.19	0.26
	(0.02)	(0.01)	(0.02)***
Former cadre	0.57	0.33	0.25
	(0.02)	(0.01)	(0.02)***
PC membership	0.28	0.15	0.13
	(0.02)	(0.01)	(0.02)***
CPPCC membership	0.26	0.36	−0.10
	(0.02)	(0.01)	(0.02)***
Firm's attributes			
Sales (million yuan)	26.72	19.25	7.47
	(2.46)	(1.22)	(2.53)***
Employment (persons)	199.58	154.36	45.22
	(12.32)	(6.99)	(13.95)***
ROA	0.19	0.16	0.03 (0.04)
	(0.05)	(0.01)	
ROE	0.27	0.23	0.04 (0.04)
	(0.05)	(0.02)	
Total assets (million yuan)	17.73	12.97	4.76
	(2.88)	(1.03)	(2.44)*
Total equity (million yuan)	13.74	10.16	3.59 (2.27)
	(2.77)	(0.93)	
Firm age	5.55	7.55	2.00
	(0.15)	(0.11)	(0.20)***
Leverage	0.26	0.20	0.06
	(0.01)	(0.01)	(0.01)***
Initial loan from banks (million yuan)	0.94	0.45	0.49
	(0.16)	(0.08)	(0.16)***
Initial loan from government institutions (million yuan)	0.15	0.02	0.13
	(0.04)	(0.01)	(0.02)***
Total initial loan (million yuan)	1.09	0.47	0.62
	(0.17)	(0.08)	(0.17)***
Total current loan (million yuan)	2.92	2.20	0.73
	(0.33)	(0.21)	(0.41)*
Tax/sales	0.06	0.06	0.00 (0.00)
	(0.002)	(0.004)	
Fees/sales	0.02	0.02	0.00 (0.00)
	(0.001)	(0.002)	
Time dealing with bureaucracies	1.79	1.89	−0.10
	(0.05)	(0.03)	(0.05)*
Trust in the legal system	0.12	0.09	0.03
	(0.01)	(0.01)	(0.01)**

Significance levels 0.1, 0.05 and 0.01 are noted by *, **, and *** respectively.

Table 3
OLS regressions examining the impacts of political connections on the performance of private firms in China

	ROA			ROE		
	(1)	(2)	(3)	(4)	(5)	(6)
Party membership	0.080**	0.084**	0.080**	0.079*	0.085**	0.081*
	(0.037)	(0.038)	(0.039)	(0.041)	(0.043)	(0.043)
Firm's attributes						
Total assets (log)	−0.062***	−0.062***	−0.063***	−0.088***	−0.088***	−0.088***
	(0.011)	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)
Leverage	−0.142**	−0.140**	−0.142**	0.224***	0.227***	0.226***
	(0.066)	(0.066)	(0.066)	(0.072)	(0.073)	(0.073)
Firm age	−0.003	−0.003	−0.003	0.000	−0.000	0.000
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Owner's human capital						
Education	0.008	0.008	0.009	0.012*	0.013*	0.013**
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Age	−0.005**	−0.005**	−0.005**	−0.006**	−0.006**	−0.006**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Management experience	0.003	0.003	0.003	0.004	0.004	0.004
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Owner's political variables						
Former public firm manager		−0.015	−0.017		−0.022	−0.023
		(0.037)	(0.038)		(0.042)	(0.042
Former cadre		−0.009	−0.008		−0.012	−0.012
		(0.034)	(0.034)		(0.038)	(0.038
PC membership			0.027			0.025
			(0.041)			(0.046
CPPCC membership			−0.010			−0.022
			(0.035)			(0.039)
Observations	2324	2324	2324	2324	2324	2324
R-squared	0.10	0.10	0.10	0.10	0.10	0.10

Standard errors are reported in parentheses.
Significance levels 0.1, 0.05 and 0.01 are noted by *, **, and *** respectively.
All regressions control for industrial and provincial dummies.

bias in the estimation. Nonetheless, the burden of solving the potential endogeneity in this context is not great, because we are mainly interested in examining whether the correlation between these variables and firm performance reduces the partial correlation between Party membership and firm performance.

In columns 4–6 of Table 3, we report the same set of regressions using ROE as the dependent variable. The effect of Party membership on ROE follows a very similar pattern to that which is shown in the first three columns of Table 3, and does not decrease when we control for other potential determinants of firm profitability.

In summary, the regressions in Table 3 consistently show that Party membership has a positive effect on firm profitability. This suggests that the Party membership dummy may not have picked up the effects of the other observable human capital and political connections variables, but is likely to have measured something about the entrepreneurs that goes beyond these variables and is valuable to business operations.

6. Further tests

In the previous section, we found that Party membership is an important determinant of firm performance. However, as this finding is consistent with both the political connections and human capital hypotheses, in this section we attempt to gain further evidence about whether political connections help private entrepreneurs to perform better.

6.1. Why is Party membership important?

We have argued that in transition economies Party membership is an important political connection or status that helps private entrepreneurs to gain access to the credit market, avoid red tape, obtain a reduction in taxes or fees, and gain better access to the legal system. We can carry out more direct tests of these predications by estimating the effects of Party membership on each of these factors.

First, we test whether political connections help a firm to gain access to the credit market, using the amount of loans from state banks and other state institutions as the dependent variable. The first three regressions that are reported in Table 4 show that Party members do indeed have an advantage in obtaining initial capital from both banks and other state institutions, with the Party membership dummy being positive and significant at the five-percent level in all three regressions. The finding that Party membership helps a firm to borrow from banks and other state institutions may partially explain why such membership affords private firms certain advantages. Previous experience as a public firm manager also helps in obtaining loans from banks for initial investment, but surprisingly being a government cadre has a negative effect.

The regression results also show that Party membership contributes positively to current borrowing, and that this positive effect is mainly driven by firm size. In column 4, we report a regression in which current bank loans is the dependent variable. The estimated coefficients of the Party membership dummy and the former public firm manager dummy both have positive coefficients, and the latter is statistically significant. However, once we control for firm size in column 5, the Party membership dummy becomes very small, which suggests that the advantage of securing current bank loans is generated by the presence in the sample of larger firms that are owned by Party members.

A potential concern is that Party members obtain more loans because in their hands assets may be more productive. If this is true, then Party members securing more loans is an efficient outcome. To examine this, we follow Banerjee and Munshi (2004) in estimating a production function in which we include an interaction term between total assets and the Party membership dummy. If assets are more productive in the hands of Party members, then this interaction term should be positive and significant. However, the regression result that is reported in the last column of Table 4 suggests that assets are no more productive in the hands of Party members, as the coefficient on the interaction term is not significant.

Second, we test whether Party membership allows firms to secure favorable treatment in terms of tax or fee

Table 4
OLS regressions examining the impacts of political connections on the access to loans by private firms in China

	Dependent variables					
	Initial loan from banks	Initial loan from government institutions	Total initial loan	Total current loan	Total current loan	Value added (log)
	(1)	(2)	(3)	(4)	(5)	
Party membership	0.422**	0.121***	0.543***	0.500	0.061	0.036
	(0.178)	(0.027)	(0.184)	(0.435)	(0.422)	(0.042)
Firm's attributes						
Employment (log)					1.843***	0.718***
					(0.146)	(0.014)
Total assets (log)						0.261***
						(0.012)
Total assets (log)×Party membership						0.012
						(0.018)
Owner's human capital						
Education	0.037	0.001	0.038	0.275***	0.163**	0.025***
	(0.026)	(0.004)	(0.027)	(0.066)	(0.064)	(0.005)
Age	−0.009	0.002	−0.006	−0.018	−0.003	−0.004**
	(0.010)	(0.001)	(0.010)	(0.025)	(0.024)	(0.002)
Management experience	0.032**	0.000	0.032**	0.106***	0.052*	0.000
	(0.014)	(0.002)	(0.014)	(0.032)	(0.031)	(0.002)
Owner's political variables						
Former public firm manager	0.502***	0.011	0.512***	0.726*	0.277	0.014
	(0.174)	(0.026)	(0.179)	(0.427)	(0.414)	(0.031)
Former cadre	−0.339**	−0.002	−0.341**	−0.252	−0.301	0.024
	(0.160)	(0.024)	(0.164)	(0.394)	(0.381)	(0.028)
Observations	2322	2322	2322	2278	2278	2263
R-squared	0.03	0.04	0.03	0.08	0.14	0.83

Standard errors are reported in parentheses.
Significance levels 0.1, 0.05 and 0.01 are noted by *, **, and *** respectively.
All regressions control for industrial and provincial dummies.

payments by estimating the tax and fee equations. We employ ordinary least-squares regressions and use two dependent variables: the tax rate, which is defined as the total tax payment divided by sales, and the fee rate, which is defined as the total fee payment divided by sales. The regression results in columns 1 and 2 of Table 5 show that entrepreneurs who are Party members do not pay higher taxes and fees, as neither of these effects is significant at the conventional levels.

Third, we examine whether Party membership reduces the time that an entrepreneur spends dealing with bureaucracy. In the third column of Table 5, we report a regression using the amount of time spent dealing with bureaucracy as the dependent variable and the same independent variables as are used in the previous regressions. We do indeed find that Party membership has a negative coefficient, but the effect is insignificant.

Finally, we find that Party membership gives private entrepreneurs more confidence in the legal system. We create the dummy variable "trust in the legal system," which indicates whether an entrepreneur would resort to the legal system in the event of a business dispute. When a private firm has a dispute in China, it can choose to seek legal redress, resort to non-formal resolution, or simply ignore the dispute. Only entrepreneurs who have confidence in the legal system would choose the legal channel. The final column of Table 5 reports the regression result with "trust in the legal system" as the dependent variable. The result shows Party members to be more likely to choose the legal method of dispute resolution, with the coefficient of Party membership being positive and significant. On average, Party members are 29 percentage points more likely to choose the legal method of dispute resolution than non-Party members.

In summary, we find some evidence that Party membership exerts a positive effect on firm performance through a number of mechanisms that are related to the institutional environment in China. In particular, we find that Party members are more likely to obtain loans from banks or other state institutions, and although they do not receive favorable treatment in terms of tax and fee payments, they do have more confidence in the legal system. These findings support the view that political connections significantly help firms to overcome market failures in a transition economy.

6.2. Where is Party membership important?

In this subsection, we examine the effect of Party membership on firm performance in regions with

Table 5
OLS regressions examining channels of the impacts of political connections on private firms in China

	Dependent variables			
	Tax/sales	Fees/sales	Time dealing with bureaucracies	Trust in the legal system
	(1)	(2)	(3)	(4)
Party membership	0.001	−0.002	−0.068	0.029*
	(0.003)	(0.003)	(0.059)	(0.015)
Firm's attributes				
Employment in log	−0.001	−0.001	0.091***	0.011**
	(0.001)	(0.001)	(0.021)	(0.005)
Owner's human capital				
Education	−0.000	−0.001**	−0.004	0.003
	(0.000)	(0.000)	(0.009)	(0.002)
Age	0.000**	−0.000	−0.013***	0.000
	(0.000)	(0.000)	(0.003)	(0.001)
Management experience	−0.000	−0.000	0.003	−0.001
	(0.000)	(0.000)	(0.004)	(0.001)
Owner's political variables				
Former public firm manager	−0.000	0.005*	0.061	−0.010
	(0.003)	(0.003)	(0.058)	(0.015
Former cadre	0.001	−0.001	0.011	−0.018
	(0.002)	(0.003)	(0.053)	(0.014
Observations	2271	2303	2161	2324
R-squared	0.04	0.05	0.06	0.03

Standard errors are reported in parentheses.
Significance levels 0.1, 0.05 and 0.01 are noted by *, **, and *** respectively.
All regressions control for industrial and provincial dummies.

different institutional environments. We test whether Party membership has a greater effect on performance in regions in which institutions are weaker by utilizing the three institutional indices that are described in Section 3. This test should help to differentiate the political connections hypothesis from the human capital hypothesis, because human capital should be more valuable (or at least no less valuable) in better institutional environments, but political connections should be more valuable in weaker institutional environments.

The regression results suggest that Party membership is more important to firm performance in provinces with less liberal markets. In Table 6, we report the ROA and ROE regressions with the interactions between these market indices and Party membership as co-variates. The institutional indices all have positive coefficients, some of which are significant at the five-percent level, which suggests that firms perform better in better institutional environments. More importantly, all of the interaction terms have negative coefficients, with those for the market development indices being significant at

Table 6
OLS regressions examining the impacts of political connections on the performance of private firms in Chinese provinces with different levels of market development and legal effectiveness

	ROA			ROE		
	(1)	(2)	(3)	(4)	(5)	(6)
Party membership	0.233***	1.856***	0.205**	0.238**	1.847***	0.168*
	(0.085)	(0.218)	(0.085)	(0.094)	(0.242)	(0.094)
Firm's attributes						
Total assets (log)	−0.061***	−0.062***	−0.061***	−0.087***	−0.088***	−0.086***
	(0.011)	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)
Leverage	−0.172***	−0.183***	−0.172**	0.195***	0.182**	0.199***
	(0.066)	(0.065)	(0.068)	(0.073)	(0.072)	(0.075)
Firm age	−0.002	−0.003	−0.003	0.001	−0.000	0.000
	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)	(0.005)
Owner's human capital						
Education	0.007	0.007	0.006	0.011*	0.011*	0.010
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)
Age	−0.005**	−0.005**	−0.005**	−0.005**	−0.005**	−0.005**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Management experience	0.002	0.003	0.003	0.003	0.003	0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Owner's political variables						
Former public firm manager	−0.004	0.002	−0.006	−0.009	−0.003	−0.011
	(0.038)	(0.038)	(0.039)	(0.042)	(0.042)	(0.043)
Former cadre	−0.003	−0.005	−0.003	−0.006	−0.007	−0.007
	(0.035)	(0.034)	(0.035)	(0.038)	(0.038)	(0.039)
Institutional Indices						
Market 1	0.280			0.300		
	(0.235)			(0.260)		
Market 1*Party	−0.997**			−1.039**		
	(0.469)			(0.518)		
Market 2		0.386**			0.490**	
		(0.192)			(0.212)	
Market 2*Party		−3.300***			−3.287***	
		(0.397)			(0.440)	
Legal effectiveness			0.001*			0.001*
			(0.000)			(0.000)
Legal effectiveness × Party			−0.001*			−0.001
			(0.001)			(0.001)
Observations	2324	2324	2265	2324	2324	2265
R-squared	0.03	0.06	0.03	0.04	0.06	0.04

Standard errors are reported in parentheses.
Significance levels 0.1, 0.05 and 0.01 are noted by *, **, and *** respectively.
The first market index is the share of employment of a province that is in the private sector. The second market index is the private share of total enterprise fixed investment.

least at the five-percent level. The negative coefficients on these interaction terms mean that Party membership becomes less valuable when the market is more developed or the legal environment stronger.

In summary, we find some evidence that Party membership exerts a larger positive effect on firm profitability in provinces with weaker markets and ineffective legal systems. These findings support the view that Party membership is an important political connection that provides significant economic benefits that help to overcome state and market failures in a transition economy.

7. Conclusion

This paper explores the positive role of personal affiliation with the ruling Communist Party in the operation of private businesses in China using a large and representative sample of private firms. We find that political connections have a positive effect on firm performance, a result that is robust to a series of specification tests. We also find that political connections play a positive role through a number of mechanisms that are related to the weak institutional environment in China. In particular, we find that Party membership is conducive to obtaining loans from banks or other state institutions, which gives enterprises that are run by Party members a distinct advantage. We also find that politically connected firms have more confidence in the legal system than their less well-connected counterparts. These findings support the view that political connections are conducive to business success in a transition economy such as that of China.

Our analysis suggests that the development of China's private sector relies mainly on mechanisms other than legal and formal institutions. As we show, one of these mechanisms is affiliation with the ruling Communist Party. However, we also show that the role of Party membership is not universally important in China, and that it is more important in regions with weak institutions. Conversely, when formal institutions are well developed, the role of Party membership diminishes. This seems to suggest that the development of formal institutions is very important in China, as in other countries.

China's market-supporting institutions are far from perfect, and are likely to remain so for some time, and thus political connections are likely to retain their importance for private firms. Although most of the private entrepreneurs in our sample joined the Party long before they started their enterprises, we expect to see more private entrepreneurs joining the Party to build up political connections in the future. The recent change in Party policy that was formalized at the Sixteenth Congress in accordance with the "three represents" concept to openly allow private entrepreneurs to join an organization that originally represented poor workers and farmers (Xinhua News Agency, November 18, 2002) will further encourage private entrepreneurs to become Party members.

Appendix A

Table A1
Definitions of variables

Variable name	Definition
Owner's attributes	
Education	Years of schooling
Party membership	A dummy variable which equals 1 if an entrepreneur has become a party member before the establishment of the private firm, and 0 otherwise
Former public firm manager	A dummy variable which equals 1 if an entrepreneur formerly served as a manager in state-owned enterprises or township and village enterprises, and 0 otherwise
Former cadre	A dummy variable which equals 1 if an entrepreneur formerly served as a government cadre, and 0 otherwise
PC membership	A dummy variable which equals 1 if an entrepreneur is a member of the People's Congress at any level, and 0 otherwise
CPPCC membership	A dummy variable which equals 1 if an entrepreneur is a member of the Chinese People's Political Consultative Conference at any level, and 0 otherwise
Firm's attributes	
Sales (million yuan)	
Employment (persons)	Number of persons employed. (take the value of 1 for a worker employed for the whole-year; 0.5 for employment for less than one year but more than 6 months; and 0.25 for employment for less than 6 months

Table A1 (*continued*)

Variable name	Definition
Firm's attributes	
ROA	Return on assets
ROE	Return on equity
Leverage	Amount of debt divided by total assets)
Initial loan from banks (million yuan)	Loan from banks and rural credit cooperatives when the firm was established
Initial loan from government institutions (million yuan)	Loan from state-owned and community-owned enterprises and institutions when the firm was established
Total initial loan (million yuan)	Sum of "initial loan from banks" and "initial loan from government institutions"
Total current loan (million yuan)	Loan balance from banks at the end of 2001
Value added (million yuan)	Value added in 2001 (After-tax profits+tax payment+wage payment)
Tax/sales	Amount of taxes divided by sales in 2001
Fee/sales	Amount of fees divided by sales in 2001
Time dealing with bureaucracies	Hours per day spent on dealing with public officials
Trust in the legal system	A dummy variable which equals 1 if the entrepreneur would choose to go to arbitration organization or court in the presence of economic disputes, and 0 otherwise
Institutional indices	
Market 1	The proportion of employment in a province that is in the private sector
Market 2	The proportion of total fixed investment in a province that comes from private enterprises
Legal effectiveness	Number of cases received by the court (weighted by provincial population)
Other control variables	
Sector indicators	14 industry sector indicators
Provincial indicators	31 provincial indicators

Appendix B

Table A2
Correlations coefficients between institutional indices (30 provinces)

	Market 1	Market 2	Legal effectiveness
Market 1	1.0000		
Market 2	0.4733	1.0000	
Legal effectiveness	0.6142	0.6911	1.0000

Appendix C

Table A3
OLS regressions examining the effect of political connections on the performance of private firms in China

	Dependent variable: ROA					
	(1)	(2)	(3)	(4)	(5)	(6)
Party membership	0.084** (0.038)	0.084** (0.038)	0.084** (0.038)	0.071* (0.039)	0.067* (0.038)	0.072* (0.039)
Firm attributes						
Employment (log)				0.019 (0.013)	0.101*** (0.017)	
Employment						0.000 (0.000)
Employment squared						-0.000 (0.000)
Total assets (log)	-0.062*** (0.011)	-0.062*** (0.011)	-0.062*** (0.011)		-0.114*** (0.014)	
Leverage	-0.140** (0.066)	-0.141** (0.066)	-0.140** (0.066)	-0.218*** (0.066)	-0.154** (0.065)	-0.208*** (0.066)
Firm age	-0.003 (0.004)	-0.003 (0.004)	-0.003 (0.004)	-0.005 (0.004)	-0.004 (0.004)	-0.005 (0.004)
Owner's human capital						
Education	0.007 (0.034)		0.008 (0.006)	-0.000 (0.006)	0.008 (0.006)	0.000 (0.006)
Education squared	0.000 (0.001)					

(continued on next page)

Table A3 (*continued*)

Owner's human capital						
High-school graduate		0.042 (0.041)				
College graduate		0.071 (0.047)				
Age	-0.005** (0.002)	-0.005** (0.002)	-0.005** (0.002)	-0.005** (0.002)	-0.004** (0.002)	-0.005** (0.002)
Management experience	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.001 (0.003)	0.002 (0.003)	0.001 (0.003)
Owner's political variables						
Former public firm manager	-0.015 (0.037)	-0.015 (0.037)	-0.015 (0.037)	-0.035 (0.038)	-0.026 (0.037)	-0.032 (0.038)
Former cadre	-0.009 (0.034)	-0.009 (0.034)	-0.009 (0.034)	-0.008 (0.034)	-0.015 (0.034)	-0.006 (0.034)
Observations	2324	2324	2324	2324	2324	2324
R-squared	0.10	0.10	0.10	0.08	0.11	0.08

The standard errors are reported in parentheses.
The significance levels 0.1, 0.05, and 0.01 are denoted by *, **, and ***, respectively.
All of the regressions control for industrial and provincial dummies.

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